



C4 Ecosystems and energy flow

C · Interaction and interdependence · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Energy enters ecosystems as light, is captured by producers through photosynthesis, and flows unidirectionally through successive trophic levels as organisms are consumed; at each transfer, only a small proportion (typically around 10%) of the energy is passed on, the rest being lost mainly as heat through respiration, and also in undigested material or unconsumed biomass. This low transfer efficiency limits the number of trophic levels an ecosystem can support and produces the characteristic pyramid shape of energy/biomass diagrams. Unlike energy, which flows through and is ultimately lost as heat, nutrients such as carbon and nitrogen are cycled and reused within ecosystems.

Worked example: follow the reasoning

A grassland ecosystem has a gross primary productivity (GPP) of $22\,000\text{ kJ m}^{-2}\text{ yr}^{-1}$, and producers use $9\,000\text{ kJ m}^{-2}\text{ yr}^{-1}$ of this in their own respiration. Step 1: net primary productivity (NPP) = $\text{GPP} - \text{respiration} = 22\,000 - 9\,000 = 13\,000\text{ kJ m}^{-2}\text{ yr}^{-1}$. Step 2: assuming a typical 10% transfer efficiency to primary consumers (herbivores), energy available to herbivores = $13\,000 \times 0.10 = 1\,300\text{ kJ m}^{-2}\text{ yr}^{-1}$. Step 3: applying the same 10% rule again to secondary consumers = $1\,300 \times 0.10 = 130\text{ kJ m}^{-2}\text{ yr}^{-1}$. Conclusion: energy available falls rapidly at each trophic transfer, which is why food chains rarely extend beyond four or five trophic levels.

Quick check · multiple choice

Why is the transfer of energy between successive trophic levels typically only around 10% efficient?

- A Because all organisms convert food into biomass with perfect efficiency
- B Because energy is lost as heat through respiration, and further energy is lost in faeces/undigested material or organisms not eaten
- C Because producers store 90% of their energy underground
- D Because energy cannot be converted between forms

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 In a lake ecosystem, phytoplankton (producers) fix $18\,000\text{ kJ m}^{-2}\text{ yr}^{-1}$ of energy (GPP), with $7\,000\text{ kJ m}^{-2}\text{ yr}^{-1}$ lost to their own respiration. Zooplankton (primary consumers) that feed on the phytoplankton assimilate energy with a transfer efficiency of 10% from the phytoplankton's net production. [6]
- (a) Calculate the net primary productivity (NPP) of the phytoplankton. [2]
 - (b) Calculate the energy available to the zooplankton (primary consumers). [2]
 - (c) Explain why the energy available to a small fish (secondary consumer) feeding on the zooplankton will be considerably less than the value calculated in (b). [2]

Final answer: _____



Answers and explanation

Question 1 [6]

(a) $\text{NPP} = \text{GPP} - \text{respiration} = 18\,000 - 7000 = 11\,000 \text{ kJ m}^{-2} \text{ yr}^{-1}$.

(b) $\text{Energy to zooplankton} = \text{NPP} \times \text{transfer efficiency} = 11\,000 \times 0.10 = 1100 \text{ kJ m}^{-2} \text{ yr}^{-1}$.

(c) Only around 10% of the energy available to the zooplankton is likely to be transferred to the fish, because most of the energy zooplankton obtain is lost as heat through their own respiration, and further energy is lost in undigested waste or in zooplankton that are not eaten by fish; consequently the fish receive only a small fraction (roughly $1100 \times 0.10 = 110 \text{ kJ m}^{-2} \text{ yr}^{-1}$) of the energy available at the previous trophic level.

Quick check answer

B. Much of the energy consumed is lost as heat during cell respiration, and additional energy is lost in undigested waste or in organisms that are never consumed, leaving only a small fraction to be incorporated into consumer biomass.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Biology HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.